BUS-F420: Equity and Fixed Income Investments

Lecture 11: Cash flows with Debt Instruments

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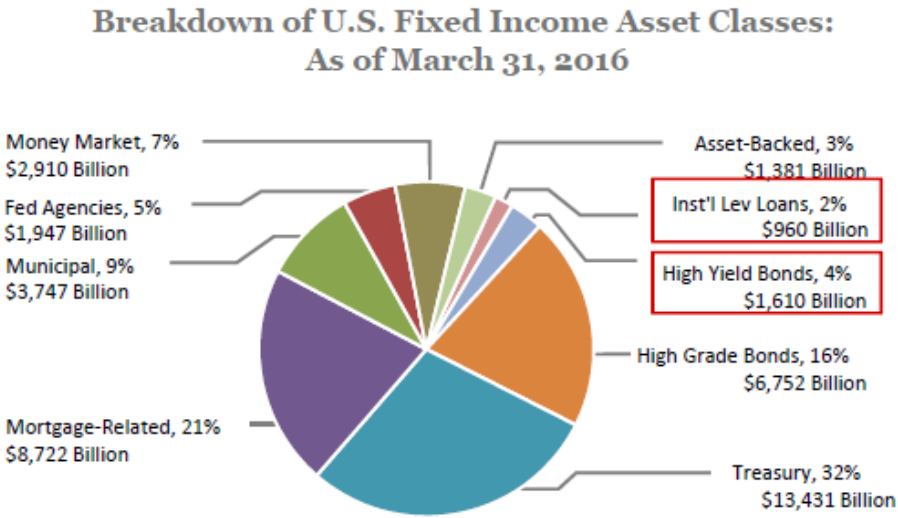
What is a fixed income asset?

Map of fixed income asset values



Roadmap for the rest of the course

- **Principles of Debt Instruments**
  - Cash flow and valuation
  - Yields and returns
- **Interest Rate Risk and the Yield Curve**
  - Measuring interest rate risk
  - The yield curve
- **Risk Management and Asset-Backed Securities**
  - Interest rate derivatives (e.g. swaps)
  - Credit risk & CDS
  - Securitization and MBS

What is the cash flow of a loan?

Fully Amortizing Loans

- Let's think about a **fixed payment** loan that **perfectly amortizes**
  - (i.e., fully paid off from fixed payments)

- Standard fixed payment schedule:

$$\underbrace{\text{Periodic Loan Payment}}_{\equiv P} = \frac{i(1+i)^n}{[(1+i)^n - 1]} \times \underbrace{\text{Loan Amount}}_{\equiv L}$$

where:

- i = periodic interest rate (annual rate / # yearly payments)
- n = number of payments (# yearly payments $\times$ # of years)

Fully Amortizing Loans

- Let's calculate the payment for an example loan with the following characteristics:
  - Annual interest rate: 9%
  - Monthly payments
  - 2 year maturity
  - Loan amount $(L) = \$100{,}000$

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  3. Calculate the loan payment:

$$P = \frac{i(1+i)^n}{[(1+i)^n - 1]} \times L = \frac{\mathbf{0.0075(1.0075)^{24}}}{\mathbf{[1.0075^{24} - 1]}} \times \mathbf{\$100{,}000 = \$4568.47}$$

Amortization Schedules

- Amortized loans come with an *amortization schedule*
  - (e.g. how much interest vs. principal do I pay each period)
- **Key idea:** As I pay off my loan, the principal that accrues interest shrinks!
  - Both *payment amount* and *maturity* account for this
- Typical amortization **front loads** interest payments
  - Q: Why?

Example of a Loan that Fully Amortizes in 6 Months

Amortization Schedule for a Fully Amortized Fixed Rate Loan

Annual interest rate	0.25
Payments per year	12
Maturity (in years)	0.5
Lockin Periods	0
Loan amount	10000
Balloon Payment	0
PV of Balloon Payment	0
Total number of payments	**6**
Periodic interest rate	**0.02**
Periodic payment	**1790.28**

Month	Balance	Payment	Interest	Amortization	New Balance
1	10000	1790	208	1582	8418
2	8418	1790	175	1615	6803
3	6803	1790	142	1649	5155
4	5155	1790	107	1683	3472
5	3472	1790	72	1718	1754
6	1754	1790	37	1754	0

Using Excel, calculate the amortization schedule for the following:

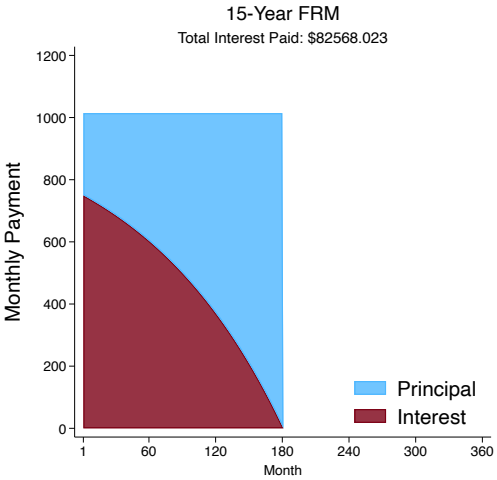
Annual interest rate	10%
Payments per year	12
Maturity (in years)	2
Loan amount	\$200,000
Total # of payments (n):	
Periodic interest rate (i):	
Periodic payment (P):	

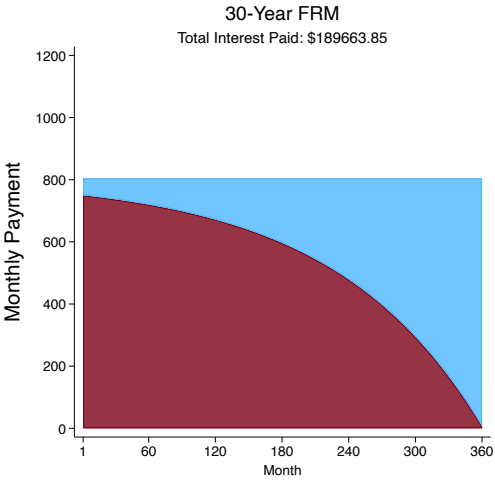
- Step 1: Calculate # payments, i, and P
- Step 2: Fill "payment" column with P
- Step 3: In month 1, set Interest = $i \times$ Balance
- Step 4: In month 1, set Amortization = Payment - Interest
- Step 5: In month 1, set New Balance = Balance - Amortization
- Step 6: In month 2, set Balance = month 1 New Balance
- Step 7: Fill payment-new balance columns in month 2, then fill all

Fixed Rate Mortgages

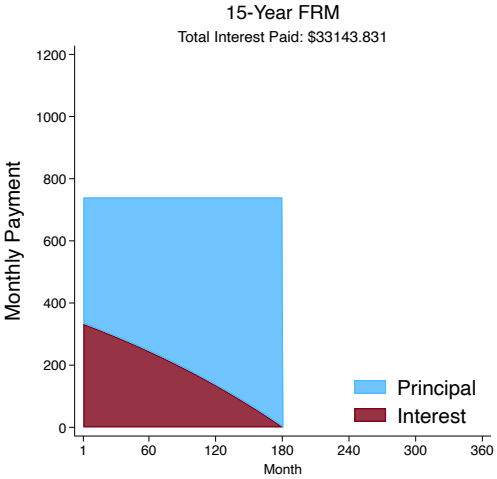
- A common example of a fully amortizing loan is a fixed rate mortgage
  - $\sim$ \$12 trillion outstanding in FRM debt
- Two common maturities
  - 15 years
  - 30 years
- **But: dramatically different amortization schedules!**
  - Q: Why?

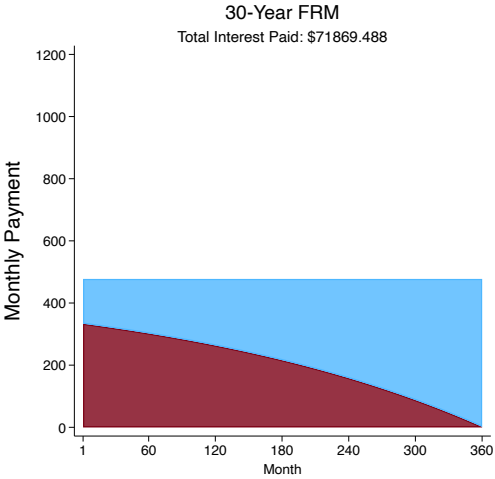
Amortization Schedule – 15 vs. 30 year FRMs ($i = 0.09$)





Amortization Schedule – 15 vs. 30 year FRMs ($i = 0.04$)





Balloon Loans

- Residential mortgages are common examples of **fully amortizing loans**
- **Commercial** mortgages not so much — often structured as **balloon loans**
  - Balloon Payment: large, fixed payment at the end of the loan
- Q: Why might this be the case?

What is the amortization schedule on a Balloon Loan?

- First, we need to establish *what needs to be amortized*!

- Split the value of the loan into two parts:

$$\underbrace{\text{Loan Value}}_{\equiv L} = \underbrace{\begin{array}{c}\text{Amount to be}\\ \text{Amortized}\end{array}}_{= L - B/(1+i)^n} + \underbrace{\begin{array}{c}\text{Present Value of}\\ \text{Balloon Payment}\end{array}}_{= B/(1+i)^n}$$

- Amoritization schedule is then the same as before, using amortization amount instead of total loan value

Example of Amortization Schedule of a Balloon Loan

Amortization Schedule for a Balloon Loan	
Annual interest rate	0.25
Payments per year	12
Maturity (in years)	0.5
Lockin periods	0
Loan amount	10000
Balloon payment	4000
PV of balloon payment	3534.52
Total number of payments	6
Periodic interest rate	0.02
Periodic payment	1157.50

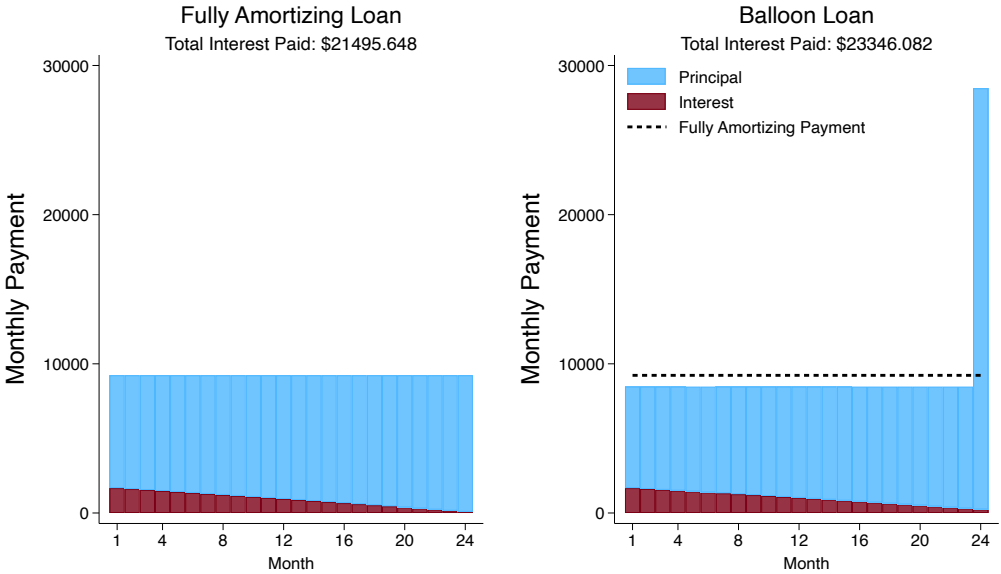
Month	Balance	Payment	Interest	Amortization	New Balance
1	10000	1158	208	949	9051
2	9051	1158	189	969	8082
3	8082	1158	168	989	7093
4	7093	1158	148	1010	6083
5	6083	1158	127	1031	5052
6	5052	1158	105	1052	4000

Your turn! Calculate the Amortization Schedule for the Following Loan

Annual interest rate	10%
Payments per year	12
Maturity (in years)	2
Loan amount	\$200,000
Balloon payment	\$50,000
PV of balloon	
Amortization amount	
Total # of payments (n):	24
Periodic interest rate (i):	0.83%
Periodic payment (P):	

- Step 1: Calculate PV of balloon payment + amortization amount
- Step 2: Calculate periodic payment
- Step 3: Repeat steps for previous example using new periodic payment

Check: Is your final "new balance" cell equal to your balloon payment?


Fully Amortizing Loan
Total Interest Paid: $21495.648
Monthly Payment
30000
20000
10000
0
1
4
8
12
16
20
24
Month
Balloon Loan
Total Interest Paid: $23346.082
Principal
Interest
Fully Amortizing Payment
Monthly Payment
30000
20000
10000
0
1
4
8
12
16
20
24
Month

Interest Only Periods

- Let's think about the characteristics of the two loans so far
  - **Fully Amortizing:** principal payments start immediately, last length of the loan
  - **Balloon:** principal payments start immediately, but are reduced due to balloon payment

- Can also in principle delay the start of amortization
  - Ex: 24 month loan, only amortize for last 12 months
  - Called "lock-in period"

- These are called **interest-only loans**
  - Common in real estate financing (ARMs, construction loans, etc.)
  - Q: Why might these be useful?

What is the amortization schedule of an interest-only loan?

- **Difference from before:** Time to *pay off principal*
  - Fully amortizing: n_{FA}
  - Interest-only: $n_{IO} = n_{FA} - k$, with k the number of lockin periods

- The payment along the life of the loan is therefore

$$P = \begin{cases} i \times L, & \text{if month} \leq k \\ \dfrac{i(1+i)^{n-k}}{[(1+i)^{n-k} - 1]} \times L, & \text{if month} > k \end{cases}$$

- **Note:** Loan *still fully amortizes*, but makes future payments *larger*

Example of an Amortization Schedule of an Interest-Only Loan

Amortization Schedule for an Interest-Only Loan	
Annual interest rate	0.25
Payments per year	12
Maturity (in years)	0.5
Lockin periods	3
Loan amount	10000
Balloon payment	0
PV of balloon payment	0
Total number of payments	6
Periodic interest rate	0.020833333
Periodic payment	3473.176751

Month	Balance	Payment	Interest	Amortization	New Balance
1	10000	208	208	0	10000
2	10000	208	208	0	10000
3	10000	208	208	0	10000
4	10000	3473	208	3265	6735
5	6735	3473	140	3333	3402
6	3402	3473	71	3402	0

Your turn!

Annual interest rate	10%
Payments per year	12
Maturity (in years)	2
Loan amount	\$200,000
Lockin Periods	8
Total # of payments (n):	24
Periodic interest rate (i):	0.83%
Interest only payment:	
Amort. p'd payment (P):	

- Step 1: Calculate the interest only payment + amortization period payment
- Step 2: Fill first 8 months with only interest payments
- Step 3: Follow steps for fully amortizing loan starting in month 9

Cash Flow Disruptions

- Two key types of risk for (fixed rate) loans:
  1. Default risk
  2. Prepayment risk

- First is obvious: many loans are *highly levered*

- Second creates a source of *interest rate risk*
  - If interest rates fall, refinance $\rightarrow$ replace with lower cash flows
  - This partially explains why amortization front-loads interest

However, Prepayment Risk $\neq$ Interest Rate Risk

- Many other reasons why a loan may be disrupted
  1. Homeowner moves cities
  2. Housing appreciation

- Many more sophisticated models of mortgage rates incorporate prepayment
  - Q: How might prepayment risk affect the payment of a mortgage?
  - We won't calculate this but useful to understand the economics

Adjustable vs. Fixed Rate Loans

- Many mortgage loans around the world are actually "adjustable" or "floating"
  - Adjustable rate = infrequent,
  - Floating = frequent

- **Key difference:** Who bears interest rate risk?

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- **Key difference:** Who bears interest rate risk?
  - FRM: banks, since mortgage values change
  - ARM: homeowners, since rates are passed through
  - Large debates on how mortgage design affects financial stability at the macro level

Bonds

Suppose the lockin period = the life of the loan. What is this?

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- This is, essentially, a fixed-coupon bond!
  - Also known as *bullet bonds* or *straight bonds*

- **Key features**
  - Repayment only occurs at maturity
  - Coupon (payment) rate is fixed

- Because there's no amortization, calculating interest payments is easy

$$P = \begin{cases} i \times L, & \text{before maturity} \\ i \times L + L, & \text{at maturity} \end{cases}$$

Zero Coupon Bonds

- Not all bonds are structured as straight bonds
- **Treasuries** are often structured as **zero coupon bonds**
  - Single repayment of principal at maturity date
  - Q: Why buy a treasury then?

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- Q: How could one earn income from a zero coupon bond?
  - A: Buy at low price, hold for some time, sell at higher price
  - We'll see this on Wednesday

Embeddable Bonds

- Think about an adjustable rate mortgage (ARM). Structured as:
  - Initial rate (e.g. 3% for 24 months)
  - Market rate (floating for remaining maturity of loan)

- Imagine if the owner of the mortgage (e.g. a bank) could *choose* to make rate floating
  - Q: What type of financial instrument is this akin to?

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- At the 24 month mark, when would the bank choose to exercise the option?

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- At the 24 month mark, when would the bank choose to exercise the option?
  - A: if market rate $>$ fixed rate

Types of Embedded Options

- **Callable and putable bonds**
  - Gives a price in which issuer to retire/payoff the bond
  - Makes cash flows **uncertain** — valuation more difficult
- **Convertible bonds**
  - Gives bondholder right to convert to equity
  - Useful if issuer is growing!
  - However, these are often callable — even more nuance to pricing
- **Exchangeable bonds**
  - Similar to convertible bonds, but with another firm's equity

Linked Bonds

- Remember that bonds return a **nominal** amount each payment period

- Two key sources of risk (among others):
  - Inflation risk (*real* return $\neq$ nominal return)
  - Currency risk (when payments are in a different currency than yours)

- Many bonds pay out cash flows in a way to removes these risks
  - TIPS — inflation protected treasuries
  - Dual-currency issues

Next time: How do we value a bond?

- **Today:** What do the cash flows look like for debt instruments?
  - Depends on amortization schedule
  - Depends on embedded options or links
  - Depends on coupon rates

- **Next time:** What is the **value** of a debt instrument?